

SIMULATION REPORT

V-SPACE FPGA Build Challenge 2026

Experiments 2–5 — Simulation Waveforms and Verification

This report contains the final Questa simulation evidence for Projects 5, 4, 3 and 2, presented in that order. Each section includes the simulation setup, verification sequence, expected functional behavior, and the corresponding waveform screenshot.

Experiment 5 — FPGA-Based Conveyor Belt Inspection System

RTL module: Conveyor_Inspection **Testbench:** tb_Conveyor_Inspection **Simulation duration:** 80 ns

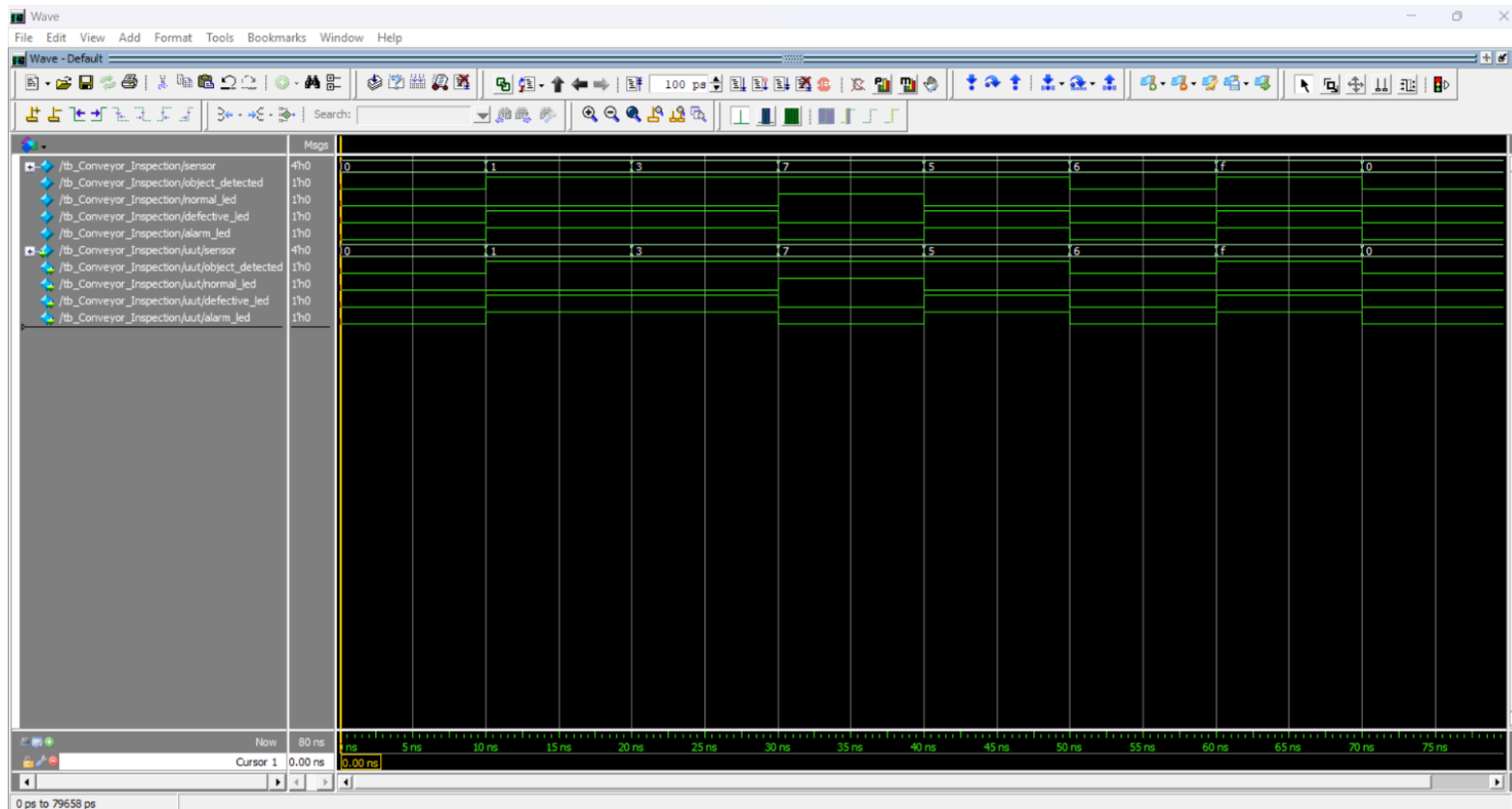
Simulation Objective

The simulation verifies object detection and distinguishes normal and defective conveyor-belt objects. A normal object is represented by sensor = 0111; defective cases activate the defect and alarm outputs when an object is detected.

Verification Sequence

Time	Input	Condition	Out-1	Out-2	Out-3	Out-4
0–10 ns	0000	No object	0	0	0	0
10–20 ns	0001	Defective object	1	0	1	1
20–30 ns	0011	Defective object	1	0	1	1
30–40 ns	0111	Normal object	1	1	0	0
40–50 ns	0101	Defective object	1	0	1	1
50–60 ns	0110	No detected object	0	0	0	0
60–70 ns	1111	Defective object	1	0	1	1
70–80 ns	0000	No object	0	0	0	0

Waveform Evidence — The screenshot below is the captured Questa waveform for this experiment.



Result: The waveform completes the 80 ns test sequence and demonstrates the expected logical transitions for sensor, object_detected, normal_led, defective_led, alarm_led. The simulation stops normally at the end of the testbench.

Conclusion

Simulation verification completed successfully with the DUT and corresponding testbench loaded in Questa. The observed transitions are consistent with the intended RTL logic.